

Personalized Movie Recommendation Engine

Netflix Prize Dataset · Collaborative Filtering at Scale · Neural Architecture Comparison

EDA & Data Pipeline

GMF vs Deep NCF

RMSE & MAP@10 Results

The Recommendation Challenge

100M+

Total ratings

480K users · 17,770 movies

98%+

Raw sparsity

Most users rate <1% of movies

45.2M

Filtered ratings

54K users · 7,127 movies

Power-Law User Activity

Median user rated 96 movies. Top 1% of users rated up to 1,390 movies — a 14× spread revealing highly skewed participation.

Movie Popularity Skew

Top 10 blockbusters each received ~50,000 ratings. Thousands of niche titles received fewer than 10.

01**Load Raw Data**

Stream 100M rows
from 4 .txt files

02**Filter**

Users ≥ 500 ratings
Movies ≥ 1000 ratings

03**Save Parquet**

45.2M rows
0.3 GB optimized

04**Train Model**

Latent Factor
Embeddings

Why Not Memory-Based KNN?

Computing full user-user similarity matrix = $O(N^2)$ memory



54,404 users $\rightarrow$ ~11.8 GB RAM just for the matrix



Infeasible on standard hardware

Solution: Latent Factor Models

Project users & movies into dense 64-128d embeddings



No full matrix — just two small embedding tables



Train end-to-end with gradient descent (AdamW)

Model Architecture Comparison

GMF — Generalized Matrix Factorization

Embedding size 128d (user & item)

Interaction Dot-product

Extras User bias, Item bias, Global mean

Parameters ~7.9 Million

Strength Strong inductive bias — resists overfitting on sparse data

VS

Deep NCF — Neural Collaborative Filtering

Embedding size 64d (user & item)

Interaction Concatenate → MLP

MLP layers 256 → 128 → 64 → 1, ReLU, 20% Dropout

Parameters ~4.0 Million

Strength Learns non-linear interactions — higher model capacity

Experimental Results — Rating Accuracy

GMF

0.7717

Test RMSE

Target beaten ✓

Deep NCF

0.8123

Test RMSE

Above target

RMSE Comparison (lower is better)

GMF 0.7717

Deep NCF 0.8123

Why GMF Wins

The dot-product's mathematical constraint acts as a regularizer on sparse data. The MLP's extra parameters overfit the training signal without learning generalizable patterns.

Experimental Results — Ranking Quality

MAP@10 Definition: Mean Average Precision at 10 — measures how well the model ranks relevant movies in the Top-10 list. Relevance threshold: actual rating ≥ 3.5 stars.

A higher MAP@10 means relevant movies appear closer to the top of the user's recommendation feed.

GMF

0.1673

2.25× better than NCF

Deep NCF

0.0743

Struggles to rank relevant items

MAP@10 Computation Steps

- ① Identify all unrated movies per user
- ② Score & rank via trained model
- ③ Select Top-10 predictions
- ④ Mark relevant if ground-truth ≥ 3.5
- ⑤ Compute AP@10 per user
- ⑥ Average across all users $\rightarrow$ MAP@10

Real-World Recommendation Output

1. Lord of the Rings: The Return of the King 5.0★

3. Star Wars: Episode V — The Empire Strikes Back 5.0★

5. Star Wars: Episode IV — A New Hope 4.8★

7. Lord of the Rings: The Two Towers 4.7★

9. Braveheart 4.6★

2. Lord of the Rings: The Fellowship of the Ring 5.0★

4. The Matrix 4.8★

6. Indiana Jones and the Last Crusade 4.7★

8. Gladiator 4.6★

10
.
Schindler's List 4.5★



Success

Model accurately clusters latent taste — identifies sci-fi/fantasy affinity purely from rating history, with no genre labels.



Failure — Popularity Bias

Dataset filtering (≥ 1000 ratings) excludes niche indie films. Predicted score saturation at 5.0 makes top-position tie-breaking arbitrary.

Key Insights & Future Work

Key Insight

Simplicity Wins

GMF (dot-product) outperforms Deep NCF on both RMSE and MAP@10. On sparse data, linear models generalise better than deep networks.

Key Insight

Production Ready

GMF's 128-d embeddings load directly into a FAISS ANN index — serving real-time recommendations to millions of users in milliseconds.

Future Work

Hybrid Architecture

Integrate content metadata (genres, director, cast) to solve the cold-start problem for new users and new movies.

Future Work

Temporal Dynamics

Incorporate rating timestamps with RNNs or Transformers (SASRec) to model shifting user taste over time and further reduce RMSE.